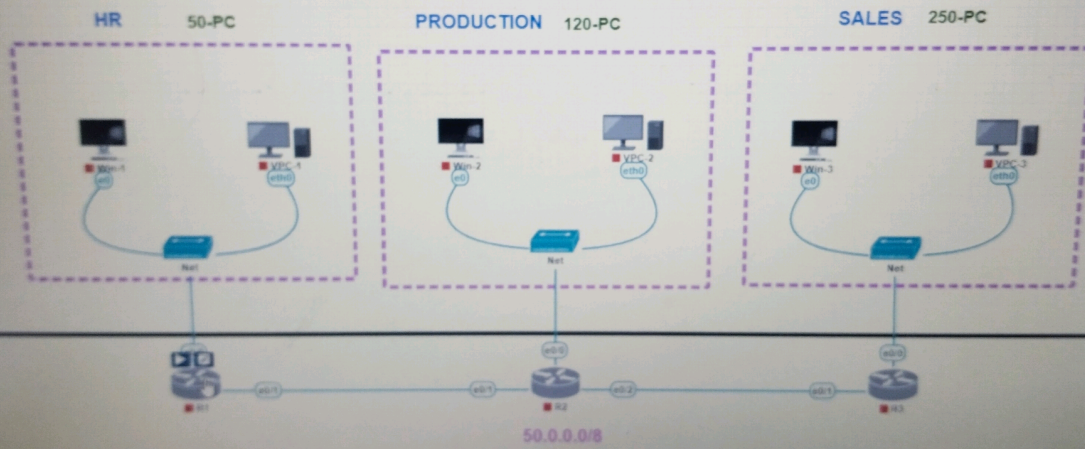


SUBNETTING



```

R1
R1(config-if)#ex
R1(config)#int e0/1
R1(config-if)#ip add 50.0.1.193 255.255.255.252
R1(config-if)#no sh
R1(config-if)#ex
R1(config)#
*Sep 10 09:17:09.416: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#ip route 50.0.1.0 255.255.255.192 50.0.1.193
%Invalid next hop address (it's this router)
R1(config)#ip route 50.0.1.0 255.255.255.128 50.0.1.194
R1(config)#ip route 50.0.0.0 255.255.255.0 50.0.1.194
R1(config)#ex
% Ambiguous command: "ex"
R1(config)#exit
R1#s
*Sep 10 09:20:34.714: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int br

```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	50.0.1.129	YES	manual	up	up
Ethernet0/1	50.0.1.193	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    50.0.0.0/24 [1/0] via 50.0.1.194
S    50.0.1.0/25 [1/0] via 50.0.1.194
C    50.0.1.128/26 is directly connected, Ethernet0/0
L    50.0.1.129/32 is directly connected, Ethernet0/0
C    50.0.1.192/30 is directly connected, Ethernet0/1
L    50.0.1.193/32 is directly connected, Ethernet0/1
R1#

```

R2

```
R2(config)#int e0/2
R2(config-if)#ip add 50.0.1.197 255.255.255.252
R2(config-if)#no sh
R2(config-if)#ex
R2(config)#
*Sep 10 09:19:26.777: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2(config)#ip route 50.0.1.128 255.255.255.192 50.0.1.193
R2(config)#ip route 50.0.0.0 255.255.255.0 50.0.1.194
%Invalid next hop address (it's this router)
R2(config)#ip route 50.0.0.0 255.255.255.0 50.0.1.198
R2(config)#exit
R2#
R2#s
*Sep 10 09:22:28.951: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int br
Interface                               IP-Address      OK? Method Status    Prot
Ethernet0/0                             50.0.1.1        YES manual up
Ethernet0/1                             50.0.1.194      YES manual up
Ethernet0/2                             50.0.1.197      YES manual up
Ethernet0/3                             unassigned      YES TFTP   down
Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
S    50.0.0.0/24 [1/0] via 50.0.1.198
C    50.0.1.0/25 is directly connected, Ethernet0/0
L    50.0.1.1/32 is directly connected, Ethernet0/0
S    50.0.1.128/26 [1/0] via 50.0.1.193
C    50.0.1.192/30 is directly connected, Ethernet0/1
L    50.0.1.194/32 is directly connected, Ethernet0/1
C    50.0.1.196/30 is directly connected, Ethernet0/2
L    50.0.1.197/32 is directly connected, Ethernet0/2
R2#
```

R3

```
R3(config)#int e0/0
R3(config-if)#ip add 50.0.0.1 255.255.255.0
R3(config-if)#no sh
R3(config-if)#ex
R3(config)#int e0/1
R3(config-if)#ip add 50.0.1.198 255.255.255.252
R3(config-if)#no sh
R3(config-if)#ex
R3(config)#
*Sep 10 09:22:07.110: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
```

```
R3(config)#
R3(config)#ip route 50.0.1.128 255.255.255.192 50.0.1.197
R3(config)#ip route 50.0.1.0 255.255.255.128 50.0.1.197
R3(config)#exit
```

```
R3#sh
*Sep 10 09:24:30.724: %SYS-5-CONFIG_I: Configured from console by console
```

```
R3#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	50.0.0.1	YES	manual	up	up
Ethernet0/1	50.0.1.198	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R3#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
50.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
C    50.0.0.0/24 is directly connected, Ethernet0/0
L    50.0.0.1/32 is directly connected, Ethernet0/0
S    50.0.1.0/25 [1/0] via 50.0.1.197
S    50.0.1.128/26 [1/0] via 50.0.1.197
C    50.0.1.196/30 is directly connected, Ethernet0/1
L    50.0.1.198/32 is directly connected, Ethernet0/1
R3#
```

QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address : 50.0.1.130
Subnet Mask : 255.255.255.192
Default Gateway : 50.0.1.129

C:\Documents and Settings\user>ping 50.0.1.2

Pinging 50.0.1.2 with 32 bytes of data:

Reply from 50.0.1.2: bytes=32 time=3ms TTL=126
Reply from 50.0.1.2: bytes=32 time=2ms TTL=126
Reply from 50.0.1.2: bytes=32 time=1ms TTL=126
Reply from 50.0.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 50.0.1.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.0.2

Pinging 50.0.0.2 with 32 bytes of data:

Reply from 50.0.0.2: bytes=32 time=3ms TTL=125
Reply from 50.0.0.2: bytes=32 time=2ms TTL=125
Reply from 50.0.0.2: bytes=32 time=1ms TTL=125
Reply from 50.0.0.2: bytes=32 time=2ms TTL=125

Ping statistics for 50.0.0.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

QEMU (Win-2)

C:\WINDOWS\system32\cmd.exe

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address. : 50.0.1.2
Subnet Mask : 255.255.255.128
Default Gateway : 50.0.1.129

C:\Documents and Settings\user>ping 50.0.0.1

Pinging 50.0.0.1 with 32 bytes of data:

Reply from 50.0.0.1: bytes=32 time<1ms TTL=254
Reply from 50.0.0.1: bytes=32 time<1ms TTL=254
Reply from 50.0.0.1: bytes=32 time<1ms TTL=254
Reply from 50.0.0.1: bytes=32 time=1ms TTL=254

Ping statistics for 50.0.0.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 50.0.1.129

Pinging 50.0.1.129 with 32 bytes of data:

Reply from 50.0.1.129: bytes=32 time=1ms TTL=254
Reply from 50.0.1.129: bytes=32 time<1ms TTL=254
Reply from 50.0.1.129: bytes=32 time<1ms TTL=254
Reply from 50.0.1.129: bytes=32 time=1ms TTL=254

Ping statistics for 50.0.1.129:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 50.0.1.130

Pinging 50.0.1.130 with 32 bytes of data:

Reply from 50.0.1.130: bytes=32 time=2ms TTL=126
Reply from 50.0.1.130: bytes=32 time=1ms TTL=126
Reply from 50.0.1.130: bytes=32 time=1ms TTL=126
Reply from 50.0.1.130: bytes=32 time=1ms TTL=126

Ping statistics for 50.0.1.130:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.0.2

Pinging 50.0.0.2 with 32 bytes of data:

Reply from 50.0.0.2: bytes=32 time=3ms TTL=126
Reply from 50.0.0.2: bytes=32 time=1ms TTL=126
Reply from 50.0.0.2: bytes=32 time=2ms TTL=126
Reply from 50.0.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 50.0.0.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

QEMU (Win-3)



C:\WINDOWS\system32\cmd.exe

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address : 50.0.0.2
Subnet Mask : 255.255.255.0
Default Gateway : 50.0.0.1

SU

C:\Documents and Settings\user>ping 50.0.1.1

Pinging 50.0.1.1 with 32 bytes of data:

Reply from 50.0.1.1: bytes=32 time=1ms TTL=254
Reply from 50.0.1.1: bytes=32 time=1ms TTL=254
Reply from 50.0.1.1: bytes=32 time=1ms TTL=254
Reply from 50.0.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 50.0.1.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.1.2

Pinging 50.0.1.2 with 32 bytes of data:

Reply from 50.0.1.2: bytes=32 time=1ms TTL=126
Reply from 50.0.1.2: bytes=32 time=2ms TTL=126
Reply from 50.0.1.2: bytes=32 time=1ms TTL=126
Reply from 50.0.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 50.0.1.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.1.129

Pinging 50.0.1.129 with 32 bytes of data:

Reply from 50.0.1.129: bytes=32 time=1ms TTL=253
Reply from 50.0.1.129: bytes=32 time=2ms TTL=253
Reply from 50.0.1.129: bytes=32 time=1ms TTL=253
Reply from 50.0.1.129: bytes=32 time=1ms TTL=253

Ping statistics for 50.0.1.129:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 50.0.1.130

Pinging 50.0.1.130 with 32 bytes of data:

Reply from 50.0.1.130: bytes=32 time=2ms TTL=125
Reply from 50.0.1.130: bytes=32 time=1ms TTL=125
Reply from 50.0.1.130: bytes=32 time=2ms TTL=125
Reply from 50.0.1.130: bytes=32 time=1ms TTL=125

Ping statistics for 50.0.1.130:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\WINDOWS\system32\cmd.exe

Start